

1 Explainability in Reinforcement Learning

2 Maxime Alaarabiou

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Chapter 1

Reinforcement Learning

This chapter introduces the fundamental concepts of Reinforcement Learning (RL), providing the necessary foundation for the ideas developed throughout the rest of this manuscript. Its structure is as follows:

Section 1.1: An overview of the milestones that led to the emergence of RL as a unified field.

Section 1.2: A formalization of the objective underlying RL algorithms.

Section 1.3: An explanation of how policies are progressively refined through interaction with the environment to approach the optimal policy.

Section 1.4: A summary of the key ideas and a discussion of the broader significance of RL.

1.1 Historical Developments

RL is today presented as a major field within artificial intelligence. However, it originates from several research traditions that progressively converged toward a common framework. In this section, we outline the main stages in the emergence of this field, highlighting the key ideas and milestones that shaped its evolution.

1.1.1 Behavioral foundations (1900–1950)

The origins of RL can be traced back to 20th-century animal psychology studies [Thorndike, 1911]. These studies aimed to understand how behaviors are formed and modified. The central idea emerging from this work is that learning is driven by the consequences of actions. Behaviors followed by positive outcomes are more likely to be repeated, whereas those associated with negative outcomes tend to gradually decrease in frequency.

This perspective was later developed within behaviorism. In this school of psychology, rewards and punishments are used to gradually shape behavior through repeated interaction with the environment [Skinner, 1938]. The term “reinforcement” in the context of learning comes from this school of thought, where it refers precisely to the process of strengthening a behavior through its consequences.

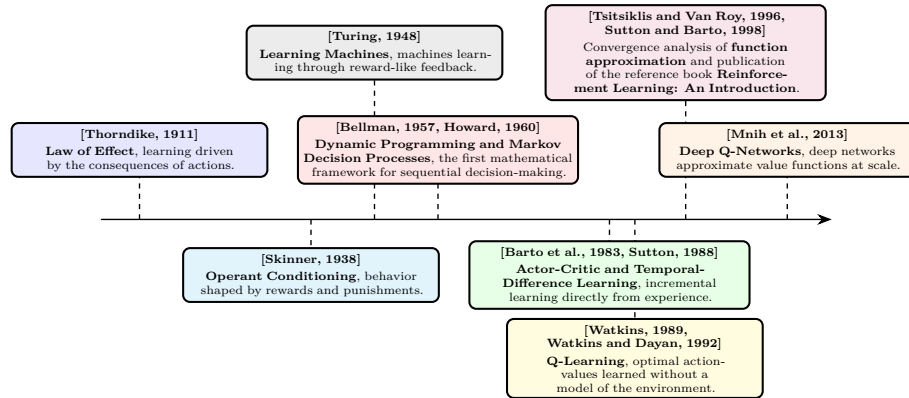


Figure 1.1: Evolution of reinforcement learning prior to its fusion with deep learning.

This idea led to the suggestion that learning principles observed in living organisms could be transferred to artificial systems. Machines were therefore envisioned as potentially learning through reward-like mechanisms, guided by evaluative feedback signals resembling a “pleasure–pain system” [Turing, 1948]. Although these studies did not involve formal mathematical models, they introduced the central idea of RL, trial-and-error learning guided by environmental feedback.

1.1.2 Sequential decision-making (1950–1970)

A second foundation of RL comes from optimal control theory. This field studies how to determine sequences of decisions that optimize a cumulative performance criterion over time. In this context, the decision-making entity is called an agent. The agent interacts with the environment by selecting actions. The function that maps an observed state to the chosen action is called a policy. The objective is to find an optimal policy, in the sense that it maximizes the expected cumulative reward over time.

This framework led to the formalization of Markov Decision Processes (MDP), which provide a mathematical model for sequential decision-making [Howard, 1960]. MDP describe a agent interacting with an environment through a set of states, actions, rewards, and transition rules. The earliest method proposed to solve MDP is dynamic programming [Bellman, 1957]. It proceeds by breaking down a complex decision problem into smaller, more manageable subproblems, solving each one, and combining the results to obtain an overall solution. However, dynamic programming requires a full and exact description of the environment, which is only feasible for small problems.

1.1.3 Emergence of reinforcement learning (1970–2000)

In order to address this issue, temporal-difference learning methods were introduced [Sutton, 1988]. This mechanism enables incremental learning directly from experience without requiring a complete model of the environment. It can be interpreted as a sample-based approximation of the Bellman equations.

Building on these ideas, the notion of value functions emerged as a way to estimate the long-term benefit of being in a given state or taking a given action. These functions assign a numerical score to states or state-action pairs, reflecting the expected cumulative reward an agent can obtain from that point onward. They thus provide a compact way to evaluate and compare different courses of action. This line of work culminated in Q-learning, which showed that optimal action-value functions can be learned directly from interaction with the environment, without requiring a model of its dynamics [Watkins, 1989, Watkins and Dayan, 1992].

The publication of the foundational reference work [Sutton and Barto, 1998] contributed to the formalization of RL as a coherent field. From that point onward, the field can be considered well established and widely recognized as a distinct area of research.

1.1.4 Consolidation of reinforcement learning (2000–2013)

A limitation of early algorithms such as Q-learning is their reliance on explicit state-action tables. Although these methods remove the need to know the full dynamics of the MDP, they still require storing a value for each possible state-action pair. As a result, the agent must effectively visit and maintain estimates for a potentially enormous number of states in the environment. This requirement becomes infeasible in large-scale MDP, where the state space is too large to be exhaustively explored or stored in memory.

To address this issue, research shifted toward more general representations of value functions and policies. Instead of maintaining explicit tables, function approximation methods were introduced. These allowed value functions and policies to be represented using parametric models such as linear approximators or perceptron-like architectures [Barto et al., 1983, Tsitsiklis and Van Roy, 1996]. This change enabled generalization across states, reducing the dependence on exhaustive exploration of the state space. The introduction of Deep Q-Networks (DQN), which demonstrated that deep Neural Network (NN) could be used to approximate value functions at scale [Mnih et al., 2013].

Figure 1.1 provides a timeline of the key milestones that shaped the development of reinforcement learning prior to its fusion with deep learning.

1.1.5 Deep reinforcement learning (2013–2026)

The introduction of Deep Learning (DL) into RL marked a major turning point in the field. A series of algorithms significantly improved scalability. Deep Deterministic Policy Gradient (DDPG) extended actor-critic methods to continuous action spaces [Lillicrap et al., 2015]. Proximal Policy Optimization (PPO) later became a standard algorithm in RL due to its simplicity, stability, and ability to perform in both continuous and discrete action spaces [Schulman et al., 2017].

These advances established a practical foundation for large-scale policy optimization. Notably, AlphaZero demonstrated that the combination of deep NN, RL, and tree search could reach superhuman performance in the game of Go [Silver et al., 2017]. This success was later extended to even more complex real-time strategy environments such as StarCraft II, where agents must handle long-term planning, partial observability, and large action spaces [Samvelyan et al., 2019]. Similarly, in Dota 2, large-scale multi-agent RL systems such as OpenAI Five

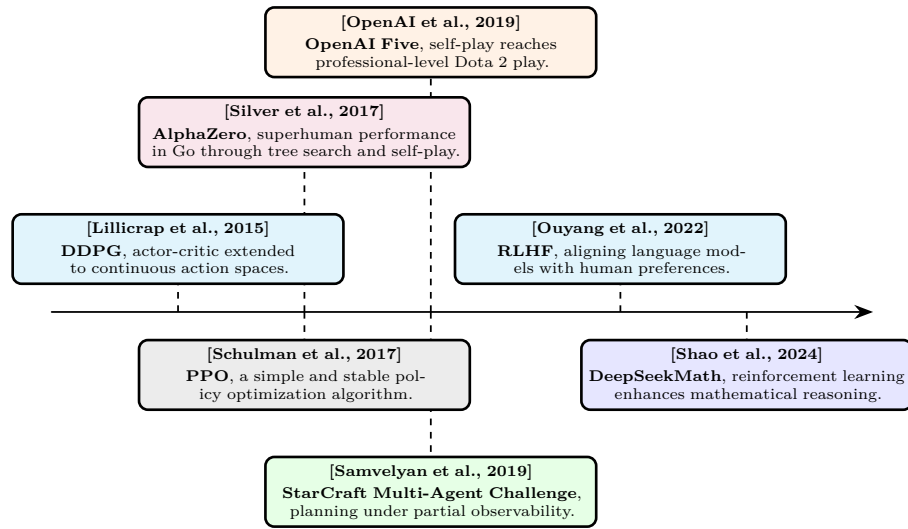


Figure 1.2: Evolution of deep reinforcement learning.

109 showed that self-play combined with deep policy optimization can achieve
 110 professional-level performance in highly dynamic and stochastic environments
 111 [OpenAI et al., 2019].

112 More recently, RL has expanded beyond its traditional application domains.
 113 With the emergence of increasingly agentic views of Large Language Model
 114 (LLM), it has become possible to train these models to perform specific tasks
 115 using RL-based techniques. In particular, RL from human feedback has become
 116 a method for aligning LLM with human preferences. It does so by formalizing
 117 the problem as an MDP and using RL to solve it [Ouyang et al., 2022]. Similarly,
 118 DeepSeekMath shows how RL can enhance the mathematical reasoning abilities
 119 of LLM [Shao et al., 2024]. It demonstrates that an artificial agent can learn
 120 to reason in natural language in order to accomplish a given task. The skills
 121 acquired during this learning process also transfer to other similar tasks, leading
 122 to improved overall performance. For instance, training a LLM with RL on
 123 mathematical problem-solving can also enhance its programming capabilities.

124 Figure 1.2 provides a timeline of the key milestones that shaped the emergence
 125 of deep reinforcement learning.

1.2 Objective

Now that the main developments shaping RL have been outlined, the formal definition of its objective can be introduced. In essence, RL aims to build a system that makes sequential decisions in order to maximize a cumulative reward signal over the long term. The purpose of this section is to translate this intuitive goal into a rigorous mathematical framework. To this end, the presentation proceeds step by step:

1. Modeling the reinforcement learning problem as a Markov Decision Process.
2. Defining the notion of a policy, which formalizes the agent's behavior.
3. Describing the interaction between the agent and the environment through trajectories.
4. Introducing value functions to evaluate policies.
5. Defining the notion of an optimal policy.

1.2.1 Markov Decision Processes

The RL framework is built around the agent paradigm, in which the decision-making entity is referred to as the agent. The agent perceives its environment, processes the information it receives, and acts upon the environment to influence future outcomes. To specify the task and the agent's capacity to perceive and act, the problem is modeled as a MDP.

Before providing a more detailed definition, it is useful to clarify the scope of the MDP problems considered in this manuscript. We therefore restrict the analysis to the following configuration:

- The single-agent learning setting is considered. If any other agents are present in the environment, they are treated as part of the environment dynamics.
- Finite-horizon MDP are considered. Each interaction sequence between the MDP and the policy has a defined beginning and a defined end. Between these two points, the number of interaction steps may vary depending on the situation, but it always remains finite.
- The partially observable framework is adopted, where the agent does not have direct access to the true state of the system, but instead relies on limited observations that provide only partial information about the environment.

The resulting framework can be formally described as a single-agent finite-horizon partially observable Markov decision process. For simplicity, we will refer to this setting as an “MDP” throughout this manuscript.

The following section aims to provide a mathematical description of the RL problem. The formulation adopted in this manuscript has been chosen to emphasize the main elements that will be used throughout the remainder of the manuscript. A MDP is defined by the tuple $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$:

- $\mathcal{S}$: The set of states, representing all possible configurations of the environment. A state belonging to this set is denoted by $s \in \mathcal{S}$. Among these, two special subsets are distinguished: initial states, from which an interaction can begin, and terminal states, in which the interaction ends. The set of initial states is denoted by $\mathcal{S}^{\text{init}}$, with an initial state written as $s^{\text{init}} \in \mathcal{S}^{\text{init}}$. The set of terminal states is denoted by $\mathcal{S}^{\text{ter}}$, with a terminal state written as $s^{\text{ter}} \in \mathcal{S}^{\text{ter}}$.
- $\mathcal{O}$: The set of observations that the agent can perceive from the environment. An observation belonging to this set is denoted by $o \in \mathcal{O}$. Unlike the true state $s \in \mathcal{S}$, which is hidden from the agent, the observation $o \in \mathcal{O}$ provides partial information about the current state.
- $\mathcal{A}$: The set of actions that the agent may execute. An action belonging to this set is denoted by $a \in \mathcal{A}$. Actions influence the dynamics of the MDP, thereby shaping its future course. They represent the agent's capacity to act upon its environment.
- $\mathcal{R}$: The set of rewards that the agent can receive from the environment. A reward belonging to this set is denoted by $r \in \mathcal{R}$. $\mathcal{R}$ is a subset of $\mathbb{R}$. Rewards can be either positive or negative, depending on whether the corresponding outcome is considered beneficial or detrimental for the agent. The agent's goal is to maximize the cumulative sum of these rewards over time.
- p : The transition function, which defines the dynamics of the environment. Given a state s and an action a , this function defines a probability distribution over the joint outcome consisting of the next state s' , the observation o , and the reward r . This function is denoted by p , and is defined as: $p : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S} \times \mathcal{O} \times \mathcal{R})$. The notation $(s', o, r) \sim p(s, a)$ means that the tuple (s', o, r) is sampled from the distribution induced by $p(\cdot \mid s, a)$. The notation $p(s', o, r \mid s, a)$ denotes the probability of observing the tuple (s', o, r) under the distribution $p(\cdot \mid s, a)$.

The two quantities, observation o and state s , are not independent. They are jointly generated via $(s', o, r) \sim p(s, a)$, which implicitly defines the relationship between them. In our MDP formalization, this relationship is left implicit because it is never manipulated directly during learning.

However, to express mathematical equations, it is convenient to introduce the conditional distribution over states given an observation. We denote this distribution by $b : \mathcal{O} \rightarrow \mathcal{P}(\mathcal{S})$. The notation $b(s \mid o)$ denotes the probability of state s given the observation o , while $s \sim b(\cdot \mid o)$ indicates that the state s is sampled from this distribution.

1.2.2 Policy

The agent's decision-making capability is modeled by a policy function. The policy defines the stochastic mapping between observations $o \in \mathcal{O}$ and actions

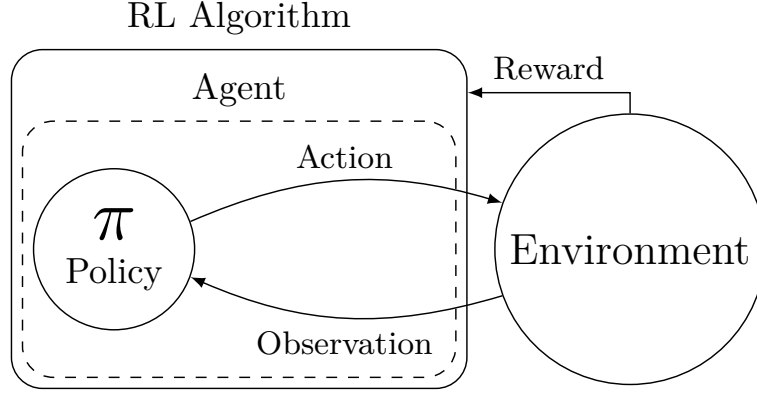


Figure 1.3: Reinforcement learning loop.

207 $a \in \mathcal{A}$. A policy function is denoted by π , and is defined as: $\pi : \mathcal{O} \rightarrow \mathcal{P}(\mathcal{A})$. The
 208 notation $a \sim \pi(\cdot \mid o)$ means that the action a is sampled from the distribution
 209 induced by $\pi(\cdot \mid o)$. The notation $\pi(a \mid o)$ denotes the probability of selecting
 210 action a under the distribution $\pi(\cdot \mid o)$.

211 It is important to note that this policy does not include any explicit mechanism
 212 for estimating the underlying state from the observations. Instead, the policy
 213 must learn from experience how to infer the information about the underlying
 214 state that is not directly available in the observations. This reflects the design
 215 of many modern RL implementations, where policies are often parameterized
 216 by architectures capable of summarizing past observations, such as recurrent or
 217 attention-based networks, without explicitly modeling the uncertainty over the
 218 underlying state.

219 Now that the MDP and the policy have both been defined, Figure 1.3
 220 illustrates how they interact within the general RL loop.

221 1.2.3 Trajectory

222 Now that both the MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$ and the policy π have been defined,
 223 the interaction between them can be characterized. A sequence of consecutive
 224 interactions between an environment and a policy is called a trajectory. A
 225 trajectory is generated through the repeated interaction between the policy and
 226 the environment, following the sequence below:

- 227 1. The environment, being in state s , provides an observation o to the policy
 228 π . Based on this observation, the policy outputs a probability distribution
 229 over actions $\pi(\cdot \mid o)$, from which an action is sampled: $a \sim \pi(\cdot \mid o)$.
- 230 2. The sampled action is executed in the environment. The environment then
 231 generates a transition: $(s', o', r) \sim p(\cdot \mid s, a)$. It moves to a new state s' ,
 232 and returns a new observation o' and a reward r .

233 A trajectory t of length T can be written as:

$$t = (s_1, o_1, a_1, r_1, s_2, o_2, a_2, r_2, \dots, s_T, o_T, a_T, r_T).$$

It is important to clearly distinguish between a trajectory distribution and its realizations. The distribution over trajectories is denoted by τ , and a particular trajectory is denoted by t . The distribution τ is determined by the initial state s , the initial observation o , the environment dynamics p , and the policy π , and is denoted by $\tau(s, o, p, \pi)$. The notation $t \sim \tau(s, o, p, \pi)$ means that the trajectory t is sampled from the distribution induced by s , o , p , and π . The notation $\tau(t \mid s, o, p, \pi)$ denotes the probability of observing the trajectory t under the distribution $\tau(\cdot \mid s, o, p, \pi)$.

An important property of the trajectory distribution is its recursive structure. Since both the policy π and the transition function p are known, the distribution over trajectories starting from s_1 with observation o_1 can be expressed in terms of the distribution over trajectories starting from the next state s_2 with observation o_2 . Specifically, the probability of a trajectory t factorizes as:

$$\tau(t \mid s_1, o_1, p, \pi) = \pi(a_1 \mid o_1) p(s_2, o_2, r_1 \mid s_1, a_1) \tau(t_2 \mid s_2, o_2, p, \pi),$$

where $t_2 = (s_2, o_2, a_2, r_2, \dots, s_T, o_T, a_T, r_T)$ denotes the trajectory starting from the second step.

There exist two special types of trajectories. First, a trajectory that terminates in a terminal state is called a terminal trajectory, and is denoted by $t_t \sim \tau_t(s, o, p, \pi)$. Second, a terminal trajectory that starts from an initial state and an initial observation is called an episode, and is denoted by $t_e \sim \tau_t(s^{\text{init}}, o, p, \pi)$.

1.2.4 Value Functions

To evaluate the ability of a policy π to collect rewards, the notion of return is introduced. The return of a trajectory $g(t, \gamma)$, is defined as the discounted sum of rewards collected along the trajectory:

$$g(t, \gamma) = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots + \gamma^{T-1} r_T = \sum_{i=1}^T \gamma^{i-1} r_i,$$

where $\gamma \in [0, 1]$ is the discount factor. This parameter controls the time horizon of the agent's objective. When γ is close to 0, the agent focuses almost exclusively on immediate rewards, leading to short-sighted behavior. As γ increases toward 1, future rewards gain more weight, and the agent adopts a more far-sighted strategy. In theory, since the goal is to maximize long-term cumulative reward, the discount factor would ideally be set to $\gamma = 1$. In practice, however, γ is often chosen slightly below 1 (e.g., 0.99) to stabilize learning, and to reflect the fact that distant future rewards are inherently more uncertain.

The return also admits a recursive form, which is central to the dynamic programming and RL algorithms presented later:

$$g(t, \gamma) = r_1 + \gamma g(t_2, \gamma),$$

From the notion of return, functions that estimate the expected return of a policy from a given starting point can be defined. These functions are called value functions. The first one considered is the state-value function V . It tells us how desirable a particular state is under a given policy π , given an observation o , the environment dynamics p , and the discount factor γ . Strictly speaking,

273 this function should be written as $V(s, o, p, \gamma, \pi)$ to reflect all its dependencies.
 274 However, since both the environment dynamics p and the discount factor γ are
 275 fixed throughout learning, they are omitted from the notation. To emphasize
 276 that the value is tied to a specific policy, π is placed as a subscript. This yields
 277 the compact notation $V_\pi : \mathcal{S} \times \mathcal{O} \rightarrow \mathbb{R}$.

278 If $V_\pi(s, o) > V_\pi(s', o')$, then, under policy π , the agent prefers being in state
 279 s with observation o over state s' with observation o' . Formally, the state-value
 280 function is defined as:

$$V_\pi(s, o) = \mathbb{E}[g(t, \gamma)], \quad \text{with } t \sim \tau_t(s, o, p, \pi).$$

281 It is worth noting that the state-value function V_π is recursive, a property
 282 that follows directly from the recursive form of the return:

$$\begin{aligned} V_\pi(s, o) &= \mathbb{E}[g(t, \gamma)], \quad \text{with } t \sim \tau_t(s, o, p, \pi) \\ &= \mathbb{E}[r_1 + \gamma g(t_2, \gamma)] \\ &= \mathbb{E}[r_1 + \gamma V_\pi(s_2)] \end{aligned}$$

283 If the expectation is expressed explicitly in terms of the distributions p and
 284 π , the Bellman equation for V_π is obtained:

$$\begin{aligned} V_\pi(s, o) &= \mathbb{E}[r_1 + \gamma V_\pi(s_2)] \\ &= \sum_{a \in \mathcal{A}} \pi(a | o) \sum_{\substack{s' \in \mathcal{S} \\ o' \in \mathcal{O} \\ r \in \mathcal{R}}} p(s', o', r | s, a) [r + \gamma V_\pi(s', o')]. \end{aligned}$$

285 Now that the value of a state has been expressed, the next step is to consider
 286 the value of taking a specific action in a given state. Indeed, it is useful to
 287 estimate the expected future gain of an action a in state s , in order to compare
 288 different actions and determine which one is the most promising under a given
 289 policy π . This function is called the state-action value function $Q_\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$.
 290 If $Q_\pi(s, a) > Q_\pi(s, a')$, then, under policy π , choosing action a in state s yields
 291 a higher expected return than choosing action a' . It is defined as:

$$Q_\pi(s, a) = \mathbb{E}[r + \gamma g(t, \gamma)], \quad \text{with } t \sim \tau_t(s', o, p, \pi), \quad (s', o, r) \sim p(\cdot | s, a).$$

292 The function Q_π can be related to V_π by recognizing that, after taking action
 293 a in state s , the remaining return from the next state s' is precisely $V_\pi(s')$. This
 294 yields:

$$\begin{aligned} Q_\pi(s, a) &= \mathbb{E}[r + \gamma g(t, \gamma)], \quad \text{with } t \sim \tau_t(s', o, p, \pi), \quad (s', o, r) \sim p(\cdot | s, a) \\ &= \mathbb{E}[r + \gamma V_\pi(s', o)] \\ &= \sum_{\substack{s' \in \mathcal{S} \\ o' \in \mathcal{O} \\ r \in \mathcal{R}}} p(s', o', r | s, a) [r + \gamma V_\pi(s', o)] \end{aligned}$$

1.2.5 Optimal Policy

The goal of RL is to find a policy that maximizes the expected cumulative reward from any observation until the end of the trajectory. This policy is called the optimal policy and is denoted by π^* . In a partially observable setting, the optimal action is the one that maximizes the expected state-action value over the distribution of possible states given the observation:

$$\pi^*(a \mid o) = \begin{cases} 1 & \text{if } a = \arg \max_{a' \in \mathcal{A}} \sum_{s \in \mathcal{S}} b(s \mid o) Q_{\pi^*}(s, a'), \\ 0 & \text{otherwise} \end{cases}$$

As shown, both V_π and Q_π can be expressed explicitly in terms of the environment dynamics p and the policy π . If p , b and π are fully known, one can theoretically compute the optimal policy by unfolding the graph of all possible trajectories starting from a given state s and observation o . This is possible precisely because V_π and Q_π are recursive functions. Solving an MDP in this way is known as dynamic programming.

In practice, most environments do have too many states to allow an explicit representation of the dynamics p . Storing and computing over the full transition function becomes intractable. To overcome this limitation, instead of directly constructing the optimal policy, an iterative approach is adopted: starting from an initial policy, it is progressively improved so that it converges toward π^* . This iterative improvement through interaction with the environment is precisely what constitutes learning in RL, and it is the subject of the next section.

1.3 Learning

Now that the optimal policy π^* has been defined, the question is how to obtain it. The recursive Bellman equations for V_π and Q_π suggest dynamic programming approaches, but these scale poorly to large state spaces. RL offers an alternative by solving the MDP iteratively and generating a sequence of policies that progressively approach π^* through direct interaction with the environment. In this section, two major families of RL methods used today are introduced: critic-only methods and actor-critic methods.

1.3.1 Critic-Only Methods

The central idea behind all critic-only approaches is to iteratively construct an approximation of the state-action value function Q_π , denoted by $\hat{Q}_\pi$. Although $\hat{Q}_\pi$ is only an approximation of the true Q_π , the policy always acts greedily with respect to this approximation.

Since $\hat{Q}_\pi$ serves as the basis for the agent's decisions, it cannot rely on more information than the agent itself has access to. It must therefore be defined solely over observations, as a function $\hat{Q}_\pi : \mathcal{O} \times \mathcal{A} \rightarrow \mathbb{R}$. Its objective is to approximate the true Q_π , which depends on the underlying states s , using only the partial information available through o . In practice, this does not necessarily pose a problem, as observations often contain sufficient information to make accurate predictions about the underlying state. Moreover, the strong representational capacity of NN allows the critic to capture these relationships without explicitly representing the mapping between observations and states.

336 This leads to the following mathematical formulation:

$$\pi(a \mid o) = \begin{cases} 1 & \text{if } a = \arg \max_{a' \in \mathcal{A}} \hat{Q}_\pi(o, a'), \\ 0 & \text{otherwise.} \end{cases} \quad (1.1)$$

337 It is precisely this way of modeling the policy that gives this family of
 338 methods its name. The policy is not represented explicitly as a standalone
 339 function. Rather, it simply emerges from searching over the set of actions to
 340 maximize $\hat{Q}_\pi$.

341 To give a clear example of critic-only methods, they are presented in their
 342 simplest form: model-free, without temporal-difference learning, without replay
 343 buffer, without bootstrapping, and using only ε -greedy exploration. Naturally,
 344 many variants and extensions of this basic scheme exist, but the core idea remains
 345 the same.

346 When implementing a RL algorithm, one of the first questions is how to
 347 represent the value function. In critic-only methods, this means choosing a
 348 model for $\hat{Q}_\pi$. Any machine learning model capable of regression task can be
 349 used for this purpose. The set of parameters of the model is denoted by θ . In
 350 deep RL, a NN is used to model this function, in which case θ represents the
 351 weights and biases of the network.

352 In our simple setting, exploration is handled via ε -greedy selection. With
 353 probability $1 - \varepsilon$, the policy selects the greedy action that maximizes $\hat{Q}_\pi$, and
 354 with probability ε it selects an action uniformly at random. Formally, this
 355 ε -greedy policy is obtained by modifying the greedy policy defined above as
 356 follows:

$$\pi(a \mid o) = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{|\mathcal{A}|} & \text{if } a = \arg \max_{a' \in \mathcal{A}} \hat{Q}_\pi(o, a'), \\ \frac{\varepsilon}{|\mathcal{A}|} & \text{otherwise.} \end{cases} \quad (1.2)$$

357 The trajectory distribution induced by the resulting policy now depends on both
 358 π and ε , and is denoted by $t \sim \tau(s, o, p, \pi, \varepsilon)$. The parameter ε controls the
 359 degree of exploration. If the policy always chooses the action it currently believes
 360 to be optimal, it may miss better alternatives that $\hat{Q}_\pi$ has not yet learned to value
 361 correctly. Exploring suboptimal actions from time to time is therefore essential
 362 to verify that $\hat{Q}_\pi$ accurately reflects the true values, while still focusing most of
 363 the time on the most promising actions. In more advanced implementations, ε
 364 can be adapted over the course of learning to control exploration dynamically,
 365 for instance by decreasing it gradually over time.

Algorithm 1: Data Collection**Input:**MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$ Policy π Discount factor $\gamma \in [0, 1]$ **Output:**Dataset $\mathcal{D}$ *Sample an initial state.*1 $i \leftarrow 1$;2 Sample $s_1^{\text{init}} \in \mathcal{S}^{\text{init}}$;3 Observe o_1 ;*Episode generation by interaction with the MDP.*4 **repeat***Action selection.*5 Sample $a_i \sim \pi(\cdot \mid o_i)$;*Environment transition.*6 Sample $(s_{i+1}, o_{i+1}, r_i) \sim p(\cdot \mid s_i, a_i)$;7 $i \leftarrow i + 1$;8 **until** $s_i \in \mathcal{S}^{\text{ter}}$;*Return computation for each step of the trajectory.*9 $T \leftarrow i - 1$;10 **for** $i = 1, \dots, T$ **do**11 $g_i \leftarrow \sum_{k=i}^T \gamma^{k-i} r_k$;*Dataset construction*12 $\mathcal{D} \leftarrow \{(s_i, o_i, a_i, g_i)\}_{i=1}^T$;13 **Return** $\mathcal{D}$

The learning process consists of two steps that repeat until convergence:

1. **Data collection.** The current policy π interacts with the MDP. For a given starting state s_1^{init} and observation o_1 , the interaction produces a episode t_e sampled from $\tau_t(s_1^{\text{init}}, o_1, p, \pi, \varepsilon)$. For each step i along the episode, the return from that step onward is $g(t_i) = \sum_{k=i}^T \gamma^{k-i} r_k$, which serves as the target value for the state-action pair (s_i, a_i) . This procedure is detailed in Algorithm 1.
2. **Model update.** The tuples $(s_i, a_i, g(t_i))$ obtained in the previous step serve as supervised training data: the state-action pair (s_i, a_i) is the input, and the return $g(t_i)$ is the associated target. These examples are used to train $\hat{Q}_\theta$. The improved approximation is denoted by $\hat{Q}'_\theta$, which now predicts returns more accurately under π . Since $\hat{Q}'_\theta$ has changed, the policy derived from it also changes. This new policy is denoted by π' .

Because the policy is now π' , the approximation $\hat{Q}'_\theta$ must be trained to predict returns under π' . This requires returning to step 1 to collect fresh interaction data generated by π' . The two steps thus alternate until the policy and the

Algorithm 2: Deep Critic-Only Reinforcement Learning**Input:**

MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$
 Architecture $\hat{Q}$
 Exploration rate $\varepsilon \in [0, 1]$
 Discount factor $\gamma \in [0, 1]$
 Learning rate $\eta > 0$
 Initial parameters θ (optional)

Output:

Trained policy π_θ

```

1 if  $\theta$  is not given then
2   | Initialize  $\theta$  randomly;
   Exploration policy as defined in Equation (1.2)
3  $\pi_\theta \leftarrow \varepsilon$ -greedy policy with respect to  $\hat{Q}_\theta$ 
   MSE loss function as defined in Equation (??)
4  $\mathcal{L} \leftarrow$  Mean Squared Error (MSE);
5 while stopping criterion not satisfied do
   Data collection see Algorithm 1
6    $\mathcal{D} \leftarrow \text{DataCollection}(\text{MDP}, \pi_\theta, \gamma)$ 
   observation-action pairs and their returns, from  $\mathcal{D}$ 
7    $\mathcal{D}' \leftarrow \{((o, a), g) \mid (s, o, a, g) \in \mathcal{D}\}$ 
   Model update with SGD see Algorithm ??
8    $\theta \leftarrow \text{SGD}(\mathcal{D}', \hat{Q}, \mathcal{L}, \eta, \theta)$ 
   Final policy without exploration as defined in Equation (1.1)
9  $\pi_\theta \leftarrow$  policy with respect to  $\hat{Q}_\theta$ ;
10 Return  $\pi_\theta$ 

```

value approximation stabilize. Algorithm 2 summarizes this whole critic-only learning loop.

Even this simple algorithm already illustrates a difficulty inherent to all forms of RL: the instability of the learning process. Each update from $\hat{Q}_\theta$ to $\hat{Q}'_\theta$ causes the policy to shift from π to π' . For the learning to remain stable, the change from $\hat{Q}_\theta$ to $\hat{Q}'_\theta$ should be as small as possible. This way, $\hat{Q}'_\theta$, trained to predict returns under π , remains reasonably accurate under π' . However, the smaller the change, the slower the learning progresses. A trade-off must therefore be struck between stability and learning speed. Many improvements introduced later in the literature aim at stabilizing this learning loop [Hessel et al., 2017].

Convergence guarantees for such algorithms have been proved under a different set of assumptions than the setting presented here. For example, in the classical convergence results, the MDP is fully observable, the value function is stored in a table, and the learning rate is decreased at a suitable rate over time [Watkins and Dayan, 1992]. In modern implementations, the value function is approximated by a NN. Theoretical convergence proofs no longer hold in this setting. Nevertheless, deep methods have demonstrated remarkable empirical

399 success, making them the tool of choice for RL tasks.

400 Critic-only approaches suffer from several limitations. The main limitation
401 is scalability to large action spaces. Since the policy must evaluate all possible
402 actions at each decision step to select the maximum, the computational cost
403 grows with the size of the action set. To address this issue, another family of
404 RL algorithms exists: actor-critic methods.

405 1.3.2 Actor-Critic Methods

406 Two principal modifications distinguish actor-critic methods from critic-only
407 approaches. First, the objective is to approximate the state-value function V ,
408 instead of the state-action value function Q . Second, the policy π is given its
409 own dedicated set of parameters. Two separate components are thus obtained:
410 a value function approximation $\hat{V}_{\theta_1}$ and a parameterized policy π_{θ_2} .

411 Analogously to critic-only methods, $\hat{V}_{\theta_1}$ is trained using supervised learning
412 on pairs obtained from interaction with the MDP. For each step i along a
413 trajectory, the input is the state-observation pair (s_i, o_i) and the target is the
414 observed return $g(t_i)$. The value function is thus trained to correct its predictions
415 based on data collected by the current policy interacting with the environment.

416 The key difference lies in how the policy π itself is optimized. Since the policy
417 now has its own set of parameters θ_2 , it can be updated directly. The update
418 uses the notion of advantage $A : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{R}$, defined as:

$$A(s, a) = g(t) - \hat{V}_{\theta_1}(s, o),$$

419 where t is the trajectory that starts by taking action a in state s and then follows
420 π for all subsequent steps.

421 When $A(s, a) > 0$, the action a performed better than expected, and the
422 policy parameters θ_2 are adjusted to increase the probability $\pi_{\theta_2}(a | o)$. When
423 $A(s, a) < 0$, the action a performed worse than expected, and the probability is
424 decreased.

425 Both $\hat{V}_{\theta_1}$ and π_{θ_2} are improved simultaneously within the same loop: the
426 critic provides a learned baseline for the actor, and the actor generates the data
427 used to train the critic. This interplay reduces the instability inherent to RL.
428 Algorithm 3 summarizes this whole actor-critic learning loop.

429 PPO (Proximal Policy Optimization), which is the current standard in RL,
430 belongs to this family of actor-critic methods. It will be used throughout this
431 manuscript for all RL experiments.

432 1.4 Conclusion

433 This chapter has provided an introduction to the fundamental principles of RL.
434 As in Chapter ??, only the essential concepts required to understand the deep
435 RL framework have been presented. The broader field of RL encompasses many
436 additional methods and extensions.

437 Despite this limited scope, the defining characteristics of RL can already
438 be observed in its simple learning procedure. A policy is iteratively improved
439 using estimates obtained through the interaction between an agent and its
440 environment. Learning therefore emerges from the repeated alternation of two

Algorithm 3: Deep Actor-Critic Reinforcement Learning**Input:**

MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$
 Architecture $\hat{V}$ (critic)
 Architecture π (actor)
 Discount factor $\gamma \in [0, 1]$
 Learning rate $\eta > 0$
 Initial parameters θ_1 of the critic (optional)
 Initial parameters θ_2 of the actor (optional)

Output:

Trained policy π_{θ_2}

```

1 if  $\theta_1$  is not given then
2   | Initialize  $\theta_1$  randomly;
3 if  $\theta_2$  is not given then
4   | Initialize  $\theta_2$  randomly;

   MSE loss function as defined in Equation (??)
5  $\mathcal{L} \leftarrow \text{MSE};$ 

6 while stopping criterion not satisfied do
   Data collection see Algorithm 1
7    $\mathcal{D} \leftarrow \text{DataCollection}(\text{MDP}, \pi_{\theta_2}, \gamma)$ 

   Advantage estimation: how much better/worse each action performed compared
   to the critic's expectation
8   for  $(s_i, o_i, a_i, g_i) \in \mathcal{D}$  do
9     |  $A(s_i, a_i) \leftarrow g_i - \hat{V}_{\theta_1}(s_i, o_i);$ 

   state and their returns, from  $\mathcal{D}$ 
10   $\mathcal{D}' \leftarrow \{(s, g) \mid (s, o, a, g) \in \mathcal{D}\}$ 

   Critic update with SGD see Algorithm ??
11   $\theta_1 \leftarrow \text{SGD}(\mathcal{D}', \hat{V}, \mathcal{L}, \eta, \theta_1)$ 

   Actor update: increase the probability of actions with positive advantage, decrease
   it for those with negative advantage
12   $\theta_2 \leftarrow \theta_2 + \eta \sum_{(s_i, o_i, a_i, g_i) \in \mathcal{D}} A(s_i, a_i) \nabla_{\theta_2} \log \pi_{\theta_2}(a_i \mid o_i);$ 

13 Return  $\pi_{\theta_2}$ 
```

441 elementary operations, which are performed until the policy and its associated
 442 value estimates converge.

443 This simple iterative structure brings RL conceptually close to DL. In both
 444 paradigms, simple operations repeated at scale can give rise to highly complex
 445 capabilities with limited human intervention. Their combination is particularly
 446 powerful. RL provides a general framework for learning through interaction,
 447 where a wide range of decision-making problems can be formulated as MDPs.
 448 DL, in turn, provides flexible function approximators capable of representing
 449 the value functions and policies required by this framework. Together, these two

450 paradigms form the foundation of deep RL, whose empirical performance has
451 led to remarkable results.

452 Originally inspired by biological learning processes, RL is now closer than ever
453 to this initial intuition. Agents learn through interaction, adapt via trial and error,
454 and progressively improve their behavior based on experience. This behavior
455 is governed by deep neural architectures, which are the closest computational
456 analog to organic brains. More than ever, RL stands as one of the core paradigms
457 for achieving artificial general intelligence [Silver et al., 2021].

458 As the capabilities of RL continue to improve, a new challenge becomes
459 increasingly important. The ability to learn complex behaviors with limited
460 human intervention can make the resulting decisions difficult to understand. To
461 address this challenge, a parallel research direction has consequently emerged.
462 This is the subject of the next chapter, which focuses on explainability in the
463 context of RL.

Acronyms

465	DDPG Deep Deterministic Policy Gradient 3
466	DL Deep Learning 3, 15
467	DQN Deep Q-Networks 3
468	LLM Large Language Model 4
469	MDP Markov Decision Processes 2–7, 10, 12–15
470	MSE Mean Squared Error 13, 15
471	NN Neural Network 3, 10, 11, 13
472	PPO Proximal Policy Optimization 3
473	RL Reinforcement Learning 1–8, 10, 11, 13–16

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